Polar H10 and Watch D2 sleep: what your data shows

Backup of 8 November 2026 · app 1.35.1 (7 Nov 2026) · sleep stage rules version 4 · 3 paired nights, 2 paired naps, 59 watch nights

96 %	45 %	0.05	+0.6 bpm
of minutes agree on asleep or awake	of minutes have the same stage	kappa: agreement beyond chance (0 = chance)	watch minus strap heart rate, 152 minutes

Main findings

1. **Both devices agree on when you slept.** Across 3 nights they matched on asleep or awake for 96 % of 940 shared minutes, and total sleep was within 11 minutes on every night.
2. **They disagree on which stage you were in.** Same stage 45 % of the time, kappa 0.05: no better than chance. The biggest gap is deep sleep: the strap finds 77 min a night on average, the watch 39 min.
3. **Heart rate agrees on average, but not minute by minute.** The watch's own minute heart rate was +0.6 bpm from the strap on average, with a typical difference of 4.9 bpm and occasional misses of up to 25 bpm. A small overall bias doesn't explain the stage disagreement, but the minute-to-minute scatter may contribute.
4. **Neither device is proven right.** Without a sleep study (brain-wave recording) there is no way to say which staging is correct. The strap's stages separate heart rate more clearly (page 6), which is expected because it uses heart rate to decide.
5. **The most useful next steps** are more paired nights, recordings that start before you lie down, and a few changes to the app listed on page 9.

Times are Singapore time. A demo night in the backup is left out. This report is a personal data review, not a medical assessment.

1. What the backup holds

Data	Count
ECG recordings	57
Polar H10 sleep recordings	8 (one demo)
Watch D2 sleep records	96
Watch minute records (heart rate, SpO_2, steps)	17569
Watch HRV readings	1325
Other watch records	3704

Sleeps recorded by both devices

Sleep	Polar H10 recording	Watch D2 in bed	Used here
4 Nov nap	10:57 · 0 h 57	11:07–11:51	Full analysis
4 Nov night	23:31 · 6 h 59	23:56–06:28	Full analysis
6 Nov night	01:53 · 4 h 39	02:03–06:28	Full analysis
6 Nov nap	copy from another device	16:04–17:52	Saved report only
8 Nov night	01:33 · 5 h 09	01:44–06:27	Full analysis

Only sleeps recorded by both devices can be compared minute by minute: 3 nights and 2 naps. The 6 Nov nap was recorded on another device, so only its saved report is in this backup. A night of 6–7 November was recorded by the watch only (the Polar H10 recording was lost before automatic saving was added in v1.35.0).

The watch on its own: 59 nights

Average per night	Watch D2
Sleep	6 h 18
Deep / light / REM	91 / 207 / 79 min
Deep sleep share	24 %
REM share	21 %
Sleep score	79
Heart rate / HRV	50 bpm / 50 ms
Breathing / SpO_2	13 per min / 97 %

2. Night by night

4 Nov night

	Polar H10	Watch D2	Combined	Watch – Polar
Total sleep	6 h 32	6 h 21	6 h 45	−11 min
Deep	1 h 39	0 h 22	1 h 51	−77 min
Light	3 h 32	4 h 29	3 h 05	+58 min
REM	1 h 12	1 h 30	1 h 49	+19 min
Asleep, stage uncertain	0 h 10	–	–	
Times woke up	1	5	3	+4
Fell asleep	23:43:31	23:56:00		+748 s
Woke up	06:30:01	06:28:00		−122 s
Heart rate (bpm)	52	50		−1.9
HRV (ms)	45	41		different methods
Breathing (per min)	12.2	14.0		+1.8
Sleep score	95	71	92	different formulas

6 Nov night

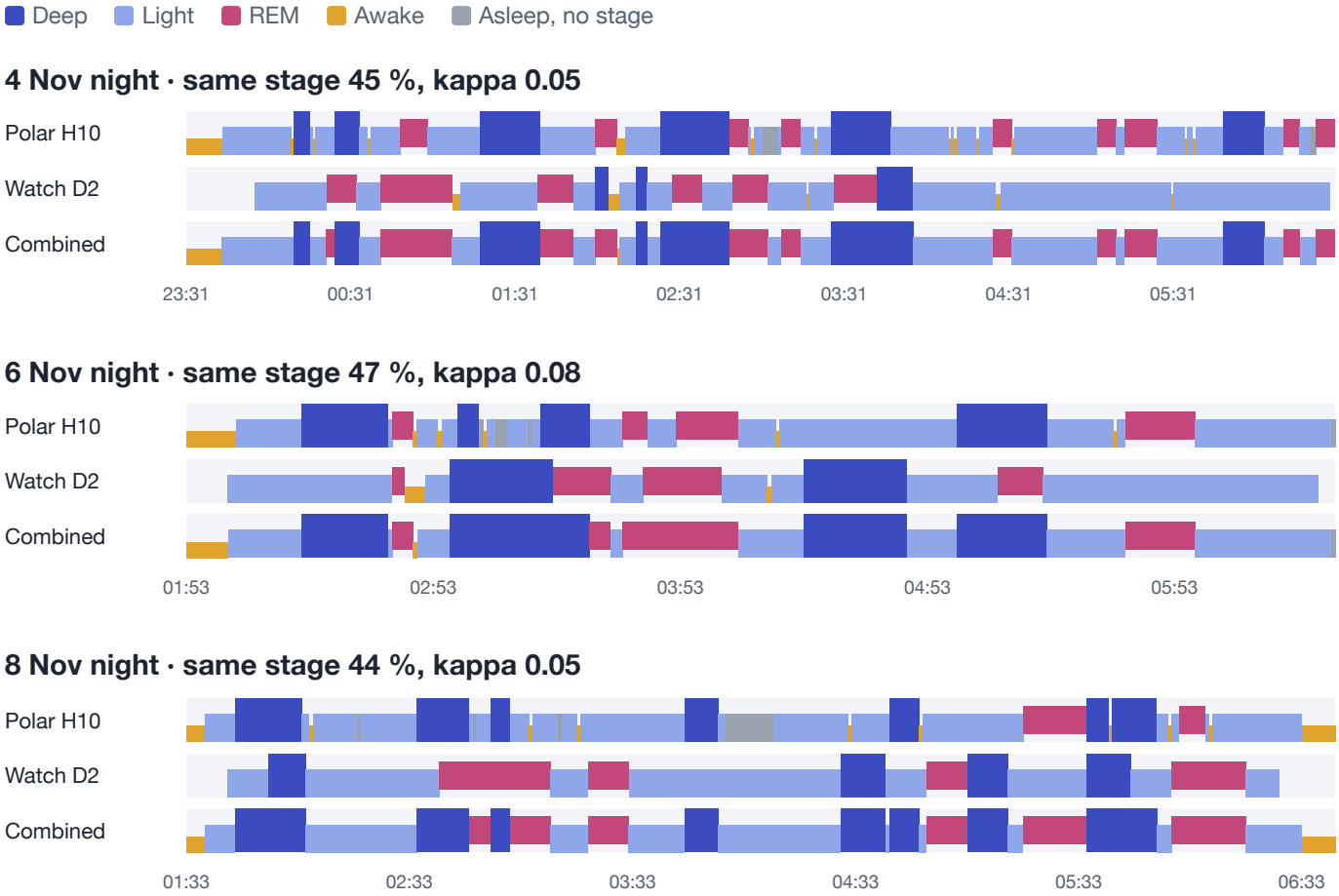
	Polar H10	Watch D2	Combined	Watch – Polar
Total sleep	4 h 23	4 h 19	4 h 28	−4 min
Deep	1 h 01	0 h 50	1 h 42	−11 min
Light	2 h 33	2 h 42	1 h 50	+9 min
REM	0 h 43	0 h 47	0 h 55	+4 min
Asleep, stage uncertain	0 h 06	–	–	
Times woke up	0	2	1	+2
Fell asleep	02:04:39	02:03:00		−100 s
Woke up	06:32:09	06:28:00		−250 s
Heart rate (bpm)	54	61		+7.1
HRV (ms)	51	48		different methods
Breathing (per min)	11.6	13.0		+1.4
Sleep score	74	63	75	different formulas

8 Nov night

	Polar H10	Watch D2	Combined	Watch – Polar
Total sleep	4 h 48	4 h 43	4 h 55	−5 min
Deep	1 h 13	0 h 45	1 h 37	−28 min
Light	2 h 54	2 h 46	2 h 02	−8 min
REM	0 h 24	1 h 12	1 h 16	+48 min
Asleep, stage uncertain	0 h 17	–	–	
Times woke up	0	0	0	+0
Fell asleep	01:37:55	01:44:00		+365 s
Woke up	06:33:25	06:27:00		−385 s
Heart rate (bpm)	55	52		−2.8
HRV (ms)	50	44		different methods
Breathing (per min)	11.9	14.0		+2.1
Sleep score	71	71	79	different formulas

Fell asleep and woke up are shown to the second; the watch reports whole minutes. The two scores use different formulas, so compare each device with its own earlier nights.

3. The nights side by side



Both devices start and end the sleep at nearly the same time, and their awake spells line up. Inside the sleep, the strap's deep sleep comes in long blocks, while the watch shows long stretches of light sleep with REM spread through the night.

Naps

4 Nov nap: Polar H10 0 h 52 asleep, watch 0 h 44. 6 Nov nap: Polar H10 1 h 55 asleep, watch 1 h 48. The watch gives naps no stages ("asleep" only), so stage agreement for naps is not available.

4. How much they agree

Night	Asleep or awake	Same stage	5-minute blocks	Kappa
4 Nov night	94 %	45 %	47 %	0.05
6 Nov night	96 %	47 %	48 %	0.08
8 Nov night	97 %	44 %	48 %	0.05
All nights	**96 %**	**45 %**		**0.05**

Where the minutes go (all 3 nights)

Polar ↓ · Watch →	Deep	Light	REM	Awake	Total
Deep	**33**	129	69	0	231
Light	71	**340**	96	8	515
REM	5	86	**38**	7	136
Awake	3	21	3	**2**	29

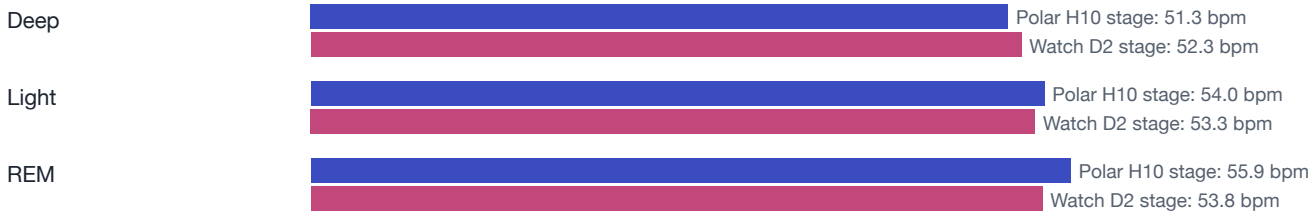
Read each row as "the strap said this", each column as "the watch said this". The darker the square, the larger the share of that row. The diagonal is agreement.

What this means

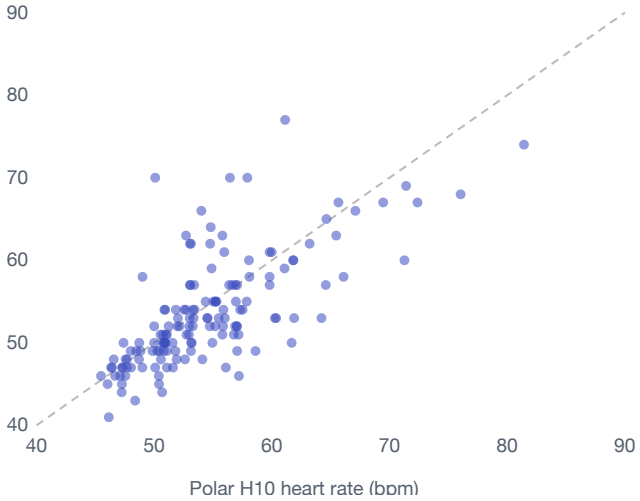
- **Sleep or awake is reliable on both.** Agreement above 90 % is in line with what wearables achieve against sleep studies for sleep and wake.
- **Deep sleep is the main disagreement.** Of 231 minutes the strap called deep, the watch called 33 deep, 129 light and 69 REM.
- **REM is the second.** The two REM labels overlap on 38 minutes only.
- **Allowing a few minutes of slack barely helps** (5-minute blocks 48 %), so this is a real difference in method, not a timing offset.

5. Heart rate during each stage

Heart rate is measured by the strap, then grouped by the stage each device gave that minute. A useful stage label should separate heart rates: lowest in deep sleep, higher and more variable in REM.



Under the strap's labels, heart rate ranges from 51.3 bpm in deep sleep to 55.9 bpm in REM. Under the watch's labels the stages are only 1.6 bpm apart. That is expected: the strap uses heart rate to decide, so it will always separate it more. It shows the strap is consistent with itself, not that it is right.



Heart rate agreement

152 minutes where the watch stored its own heart rate during a paired night.

Average difference, watch – strap	+0.6 bpm
Typical size of difference	4.9 bpm
95 % of differences between	–23.8 and +25.1 bpm
Watch HRV readings during these nights	93

Points on the dashed line mean the same heart rate on both.

6. How the night is built

	Polar H10	Watch D2	Usual pattern
Deep sleep in first / middle / last third	48 % / 29 % / 23 %	34 % / 46 % / 20 %	Mostly early
REM in first / middle / last third	11 % / 31 % / 58 %	44 % / 36 % / 20 %	Mostly late
Typical deep spell	12 min	12 min	10–40 min
Typical REM spell	7 min	13 min	10–30 min
Typical light spell	7 min	11 min	
Stage changes per night	41	20	
Fell asleep: watch – strap		+748 s, –100 s, +365 s	
Woke up: watch – strap		–122 s, –250 s, –385 s	

Reading this

- **Start and end of sleep:** close on both, typically within 12 minutes.
- **Deep sleep timing:** the usual pattern puts most deep sleep in the first third of the night. Neither device shows that clearly here; your recordings often start late (01:33, 01:53), which cuts the early night short.
- **REM timing:** the usual pattern puts most REM late. Compare the two rows above: the device closer to "mostly late" fits better on this point.
- **Spell lengths:** the strap's rules keep deep and REM spells at least 5 minutes; the watch reports in whole minutes with its own smoothing. Neither matches a sleep lab's 30-second scoring.

7. Signal quality of the Polar H10

Night	Heart and movement	Heart rate	HRV	Movement	Breathing	Stage uncertain	Possible quiet wake
4 Nov night	94 %	94 %	98 %	100 %	95 %	10 min	2
6 Nov night	93 %	93 %	98 %	100 %	95 %	6 min	0
8 Nov night	88 %	88 %	94 %	100 %	94 %	17 min	0

- **Movement is always complete**; heart data drops out for 5–10 % of the night. Gaps come from the strap losing skin contact or the Bluetooth link to the phone or laptop.
- **Uncertain minutes** are sleep with too little heart and breathing signal to name a stage. They still count as sleep.
- **Possible quiet wake** marks spells where heart rate rose without movement. It's a note only and doesn't change stages or the score.
- **Breathing** comes from chest movement and from the heartbeats, combined. Its accuracy hasn't been checked against a breathing belt.

Recording problems seen in this period

- One night (6–7 Nov) was lost because the page closed before Stop. v1.35.0 now saves every hour and restores after a crash.
- Two nights started after 01:30, so the first part of the night isn't in the comparison.

8. Suggestions for improvement

When recording

1. **Start both before you lie down** and stop after you get up, so the whole night is compared, including the deep-sleep-rich first hours.
2. **Moisten the strap's electrodes and wear it snugly** below the chest muscles, to cut the 5–10 % heart data gaps.
3. **Keep the phone or laptop plugged in, within 2 metres, with the page on screen.** Turn off Chrome's experimental setting at night.
4. **Wear the watch a finger's width above the wrist bone**, snug but comfortable, so its own heart rate stays close to the strap's.
5. **Collect at least 10 paired nights.** Three nights can show disagreement, but not which device is consistently higher or lower.
6. **Mark a late start** on the sleep report when the recording began after you fell asleep.

In the app

1. **Weight the start of the night less when the recording starts late**, as the deep-early pattern can't be judged.
2. **Show agreement over time** on Synchronised Sleep Tracking History, so a trend across nights is visible.
3. **Report deep and REM as ranges or with confidence** rather than exact minutes, as the methods differ by up to an hour a night.
4. **Test the five-minute stage rule and breathing settings** once 10 paired nights are available, judging by stability across nights, not by matching the watch.
5. **Add a "stable or unstable sleep" measure** from heart and breathing coupling, as a separate figure rather than as deep sleep.

What would settle which device is right

One or two nights in a sleep lab (polysomnography) while wearing both. Even a single night would show whether the strap's deep sleep or the watch's light sleep is closer, and could be used to tune the app's rules. Short of that, treat both devices' deep and REM figures as estimates, and rely on total sleep, waking and heart rate, where they agree well.

9. How this was worked out, and its limits

Method

- Polar H10 sleep was re-analysed from the raw recordings in the backup with the app's current rules (version 4): 30-second blocks from movement, heartbeats and breathing.
- Watch D2 stages are the watch's own, one per minute, from the night it records between falling asleep and waking.
- Both were put on the same clock minute by minute. The strap's 30-second blocks became one minute each by the most common stage.
- "Same stage" counts minutes where both named a stage. Kappa removes the agreement expected by chance from how often each stage occurs.
- Heart rate per stage uses the strap's heart rate for every minute, grouped by each device's label.

Limits

- **Small sample:** 3 paired nights, two of them starting after 01:30. Results can change with more nights.
- **No reference:** neither device was compared with a sleep study, so "agreement" is not "accuracy".
- **Different inputs:** the strap measures every heartbeat electrically and chest movement; the watch uses light through the skin, wrist movement and its own processing, which isn't published.
- **The app's rules are experimental** and were tuned partly on these same nights, which can make them look better on them than they would on new nights.
- **Watch minute heart rate** is stored only some of the time, so the heart rate comparison uses 152 minutes rather than every minute.

Words used

Kappa	Agreement beyond chance: 1 is complete, 0 is no better than chance, below 0.2 is weak.
Deep sleep	Slow-wave sleep (N3 in a sleep lab), when the body is most at rest.
REM	Rapid-eye-movement sleep, when most dreaming happens and heart rate varies.
HRV	Heart rate variability: how much the time between beats changes. Higher at rest usually means more relaxed.
Coverage	Share of the recording with usable heart and movement data.

Made by Claude (AI) from your OpenBeat backup on 8 November 2026. Not a medical device; see a doctor about any symptoms.